

Fall-26 90819 Final Project Data Source Inventory

Submitted by Keith Oxley

1) Link to dataset to be explored:

<https://docs.google.com/spreadsheets/d/1WvinnB6o1K7ffaR84NLIUpkuPfPiZrhyd6rgMNSJQoc/edit?usp=sharing>

Link to corresponding Energy & Policy Institute report the dataset is associated with:

<https://energyandpolicy.org/new-nationwide-data-reveals-utility-specific-disconnection-information/>

Feedstock for dataset are US Energy Information Administration (EIA) Form 112 submissions as detailed in the report's data processing methodology found at <https://github.com/Energy-Equity-Project/eia-112-data-pipeline>

EIA Form 112 Submission Process

- *Required Respondents: Electric and natural gas entities serving residential customers*
- *Due Date: Submit annually on or before February 28*

Modes of Submission:

- *EIA Respondent Portal (Preferred internet-based option)*
- *Secure File Transfer (Encrypted upload option for saved files)*
- *Note: Email, fax, and paper mail are not accepted.*

Form Structure

- *Schedule 1 (Identification & Services):*
 - *Part 1 covers entity name, EIA ID, primary contact, supervisor, and respondent type*
 - *Part 2 covers submission methods*
 - *Part 3 collects the number of states served and utility types provided (electricity, natural gas, or both)*
- *Schedule 2 (Monthly Data by State):*
 - *Part 4 requires monthly figures (January through December) for each state and utility type combination*
 - *Data points include total residential customers, final nonpayment notices, nonpayment disconnections, and reconnections*

2) A brief discussion of the datasets' structure and variables (note: summary generated by ChatGPT) follows on the next three pages

EIA Form 112 Data: Dataset Structure and Variables Summary

Workbook reviewed: EIA Form 112 Data (Last Updated 6/10/2026)

1. Workbook structure

The workbook contains 8 worksheets. Six are utility-level 2024 data tables, while Statistics and Rankings are summary or derived sheets. The utility tables largely share a common schema, facilitating comparisons across utility groups.

Worksheet	Rows*	Columns	Purpose
Statistics	19	4	Summary statistics comparing utility groups
Rankings	50	4	State-level 2024 disconnection statistics/rankings
2024-IOUs	169	8	Investor-owned utilities
2024-PP	1,571	20	Full public-power dataset
2024-MUNIS	444	20	Municipal electric utilities
2024-COOPS	509	20	Electric cooperatives
2024-SMALL-PP	947	20	Smaller public-power electric utilities
2024-LARGE-PP	57	20	Larger public-power electric utilities

*Rows exclude the header row. Several sheets are subsets or summaries rather than independent datasets.

2. Main public-power dataset: 2024-PP

This is the largest and most comprehensive utility-level sheet, with 1,571 observations and 20 variables. Each row appears to represent a utility's aggregated 2024 observations.

Variable	Meaning / role
state	Two-letter state abbreviation
utility_name	Name of the utility
energy_type	Utility/service type: electric or gas
year	Reporting year; all observations are 2024
ownership	Utility ownership/governance type
ownership_source	Source used to determine ownership classification
parent	Parent organization, where applicable
customer_count	Number of customers used as the denominator for rate calculations
shutoffs	Number of service shutoffs/disconnections
reconnections	Number of reconnections
final_notices	Number of final notices issued
shutoff_rate	Shutoffs relative to customer count
shutoff_rate_perc	Shutoff rate expressed as a percentage
reconnection_rate	Reconnections relative to shutoffs
final_notice_rate	Final notices relative to customer count
shutoff_rate_national_percentile	Utility's shutoff-rate percentile nationally
shutoff_rate_state_percentile	Utility's percentile within its state
shutoff_rate_ownership_percentile	Utility's percentile within its ownership category

Variable	Meaning / role
bad_data_flag	Field intended to identify problematic observations
data_quality_note	Field intended to document data-quality issues

$\text{shutoff_rate} = \text{shutoffs} / \text{customer_count}$

$\text{shutoff_rate_perc} = 100 \times \text{shutoff_rate}$

Thus, a shutoff_rate of 0.25 corresponds to approximately 25 shutoffs per 100 customers.

Composition of 2024-PP

The 1,571 observations comprise 1,004 electric observations and 567 gas observations. Ownership categories are distributed as follows:

Ownership	Observations
Municipal	949
Cooperative	534
Political Subdivision	82
State	4
Federal	2
Total	1,571

3. Public-power subset sheets

2024-MUNIS contains 444 observations, all representing 2024 municipal electric utilities. 2024-COOPS contains 509 observations, all representing 2024 cooperative electric utilities.

2024-SMALL-PP contains 947 observations and 2024-LARGE-PP contains 57 observations. Together they contain 1,004 observations, exactly matching the number of electric observations in 2024-PP. They therefore appear to partition the public-power electric observations by utility size.

The customer-count ranges indicate a division at approximately 100,000 customers: the largest customer count in 2024-SMALL-PP is 98,555, while the smallest in 2024-LARGE-PP is 100,597. This supports comparisons by both ownership and utility size.

4. Investor-owned utilities: 2024-IOUs

The IOU sheet has a simpler structure: 169 utilities and 8 variables. All observations have Ownership = Investor Owned.

Variable	Meaning
State	Utility state
Ownership	Ownership category
Utility	Utility name
Holding Company	Parent/holding company
Number of Customers	Customer base
2024 Disconnection Rate	2024 disconnection rate
Total 2024 Disconnections	Number of disconnections
Ranking (by Disconnection Rate)	Utility ranking according to disconnection rate

The IOU table does not use exactly the same schema as the public-power table. Combining them requires harmonizing fields such as Utility with utility_name, State with state, Number of Customers with customer_count, Total 2024 Disconnections with

shutoffs, and the corresponding disconnection/shutoff rate measures.

5. Rankings sheet

Rankings is a state-level aggregated table rather than a utility-level dataset. Its four variables are State, Total 2024 Disconnections, Average 2024 Disconnection Rate, and Total Number of Customers. Some rate entries contain text such as 'N/A (No Investor-Owned Utilities)', so the rate column requires cleaning before purely numeric analysis.

6. Statistics sheet

Statistics is a presentation/summary sheet rather than a conventional tidy dataset. It contains blocks reporting the number of utilities, average shutoff rate, and median shutoff rate for categories including investor-owned utilities, public power, and selected utility groups. The underlying utility-level sheets are preferable for primary analysis.

7. Data-quality and structural observations

The main 2024-PP table is highly complete for its core quantitative variables. State, utility name, energy type, year, ownership, customer count, shutoffs, reconnections, final notices, and the major rate/percentile fields have essentially no missing observations.

The parent field is missing for 1,570 of 1,571 observations. `bad_data_flag` and `data_quality_note` are entirely blank in the current 2024-PP sheet, while `reconnection_rate` has one missing value.

`customer_count` is stored as a continuous/decimal value rather than necessarily as an integer. Values such as 46885.166667 suggest that the field may represent an average customer count over a reporting period rather than a simple year-end integer count. This interpretation should be confirmed against the source documentation before rates are interpreted.

Overall interpretation

The workbook is fundamentally structured to study utility service disconnections in 2024, especially differences associated with utility ownership and size. Its analytical flow can be summarized as:

Utility characteristics -> customers -> notices/shutoffs/reconnections -> rates and percentiles

The most useful starting table for analysis is 2024-PP, with 2024-MUNIS, 2024-COOPS, 2024-SMALL-PP, and 2024-LARGE-PP functioning primarily as convenient subsets. The 2024-IOUs table provides the corresponding investor-owned comparison group, but requires variable harmonization before direct combination with public-power data.